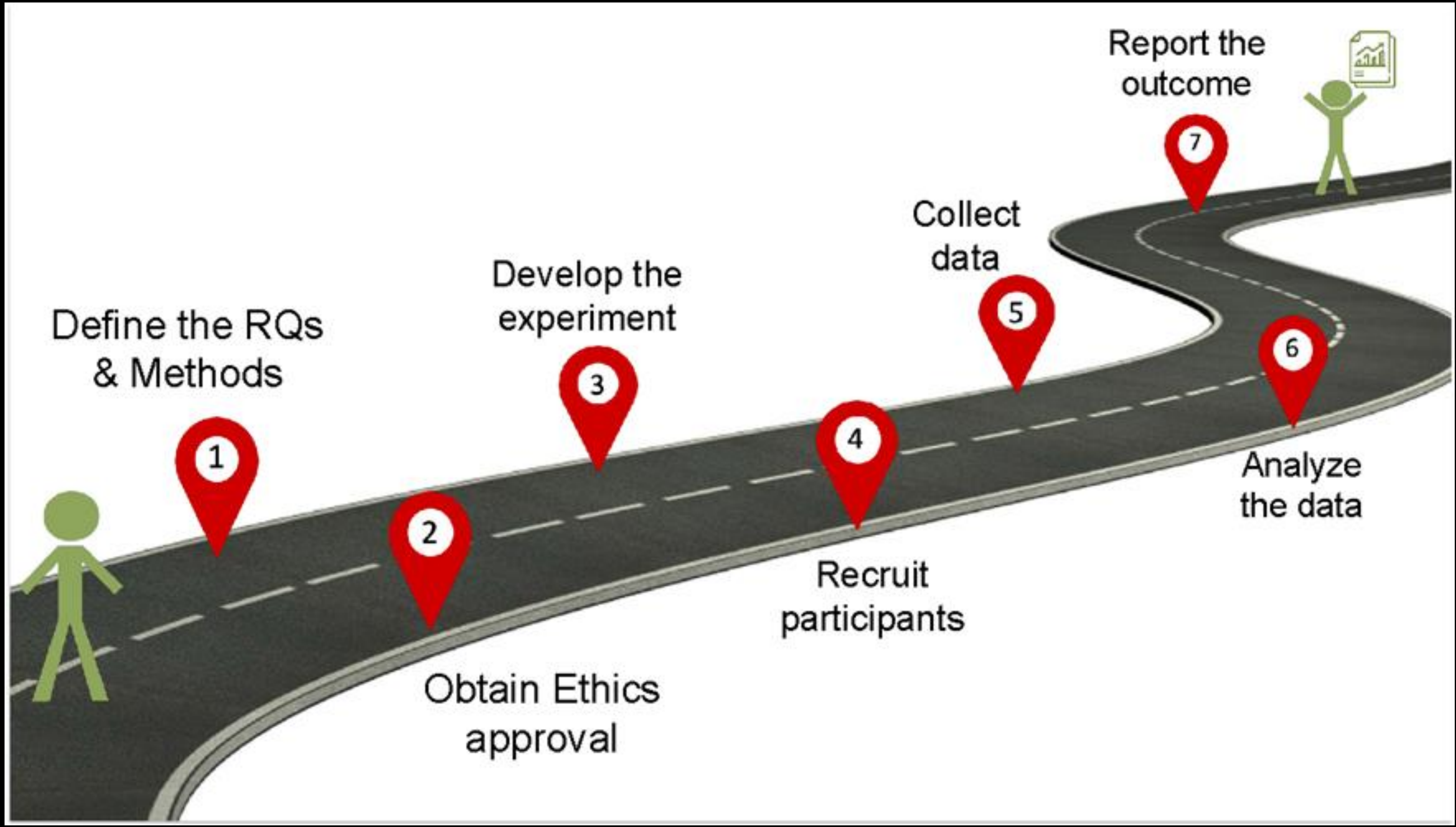
Designing HCI Experiments

Define your methodology



Ethical approval

Since HCI research involves humans, researchers must respect the safety, welfare, privacy and dignity of human participants in their research and treat them equally and fairly.

The approval process is normally governed by the institution where the research is conducted (Institutional Review Board (IRB), Ethics Review Committee (ERC), etc.)

Ethics guidelines

Participants have the right to know:

- The nature of the research (hypotheses, goals and objectives, etc.)
- The research methodology (e.g., medical procedures, questionnaires, participant observation, etc.)
- Any risks or benefits
- The right not to participate, not to answer any questions, and/or to terminate participation at any time without penalty
- The right to anonymity and confidentiality

Define your methodology

The term *method* or *methodology* refers to the way an experiment is designed and carried out.

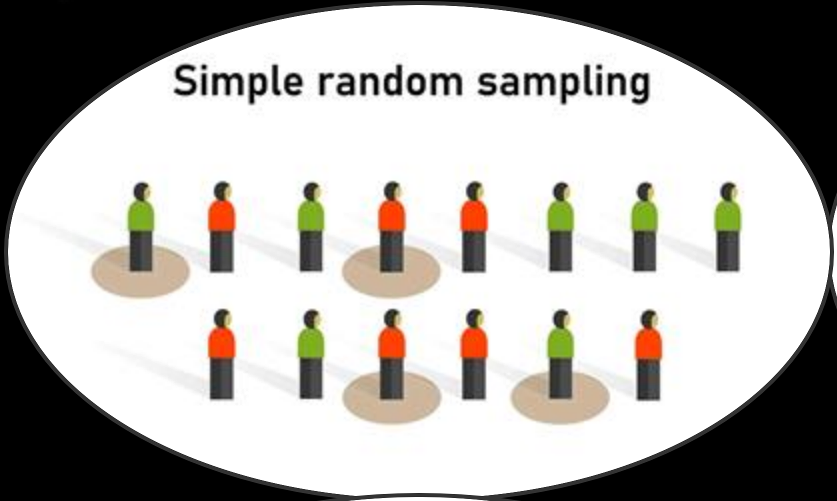
- Participants
- Materials or apparatus (hardware and software)
- User tasks
- Experimental design and conditions (IV)
- Procedure (the order of tasks/conditions)
- Evaluation (DV)
- Data analysis

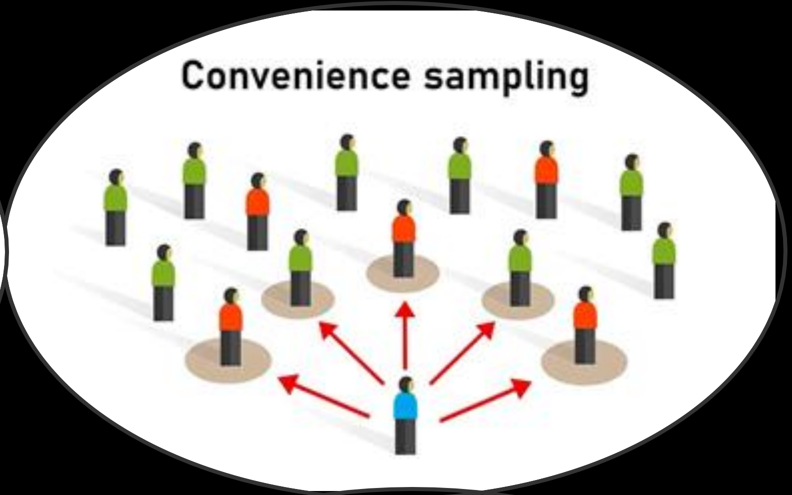
Participants

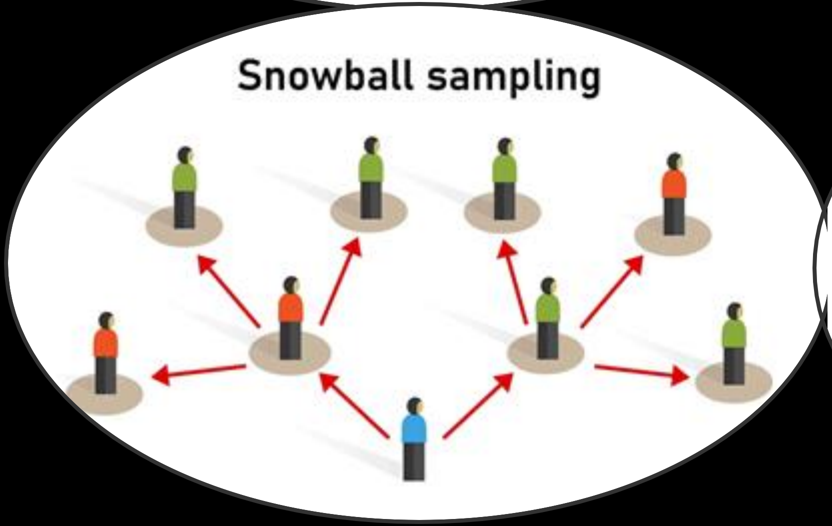
- We cannot test the whole population
- Instead, we can take some people from the population as *sample*
- Samples should reflect the population

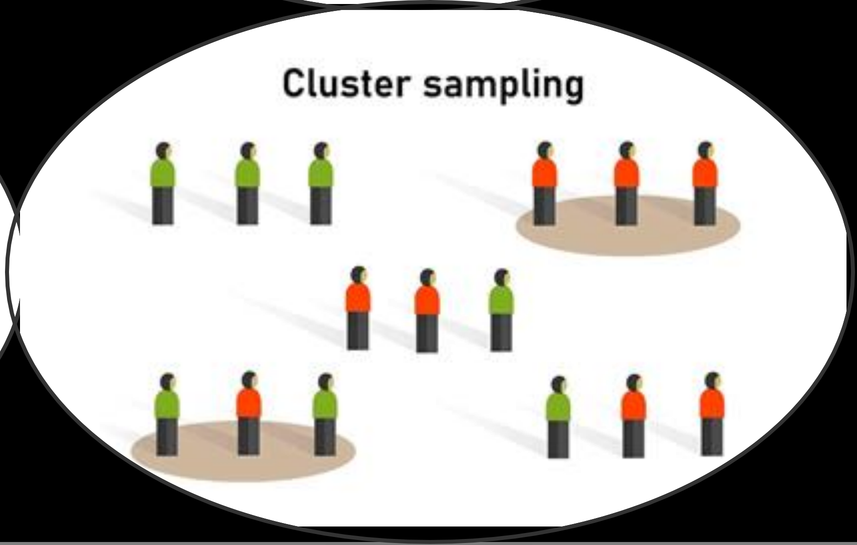
Choosing participants is called sampling

Sampling methods







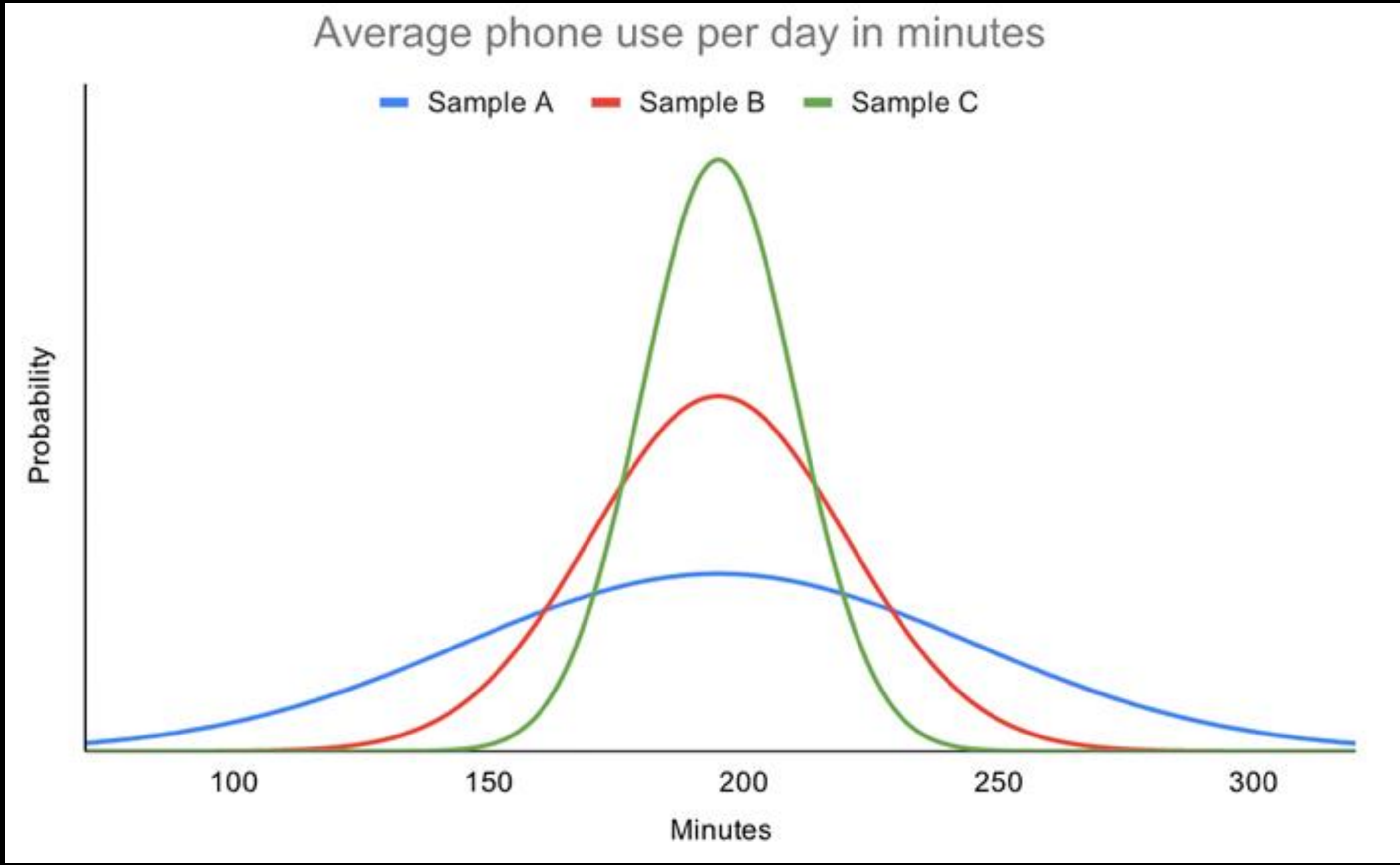


How many participants do I need?

- The performance varies between humans
- More participants -> better results (higher statistical validity)
- More participants -> higher costs
- For you research project recruit 10-12 participants per condition

How many participants do I need?

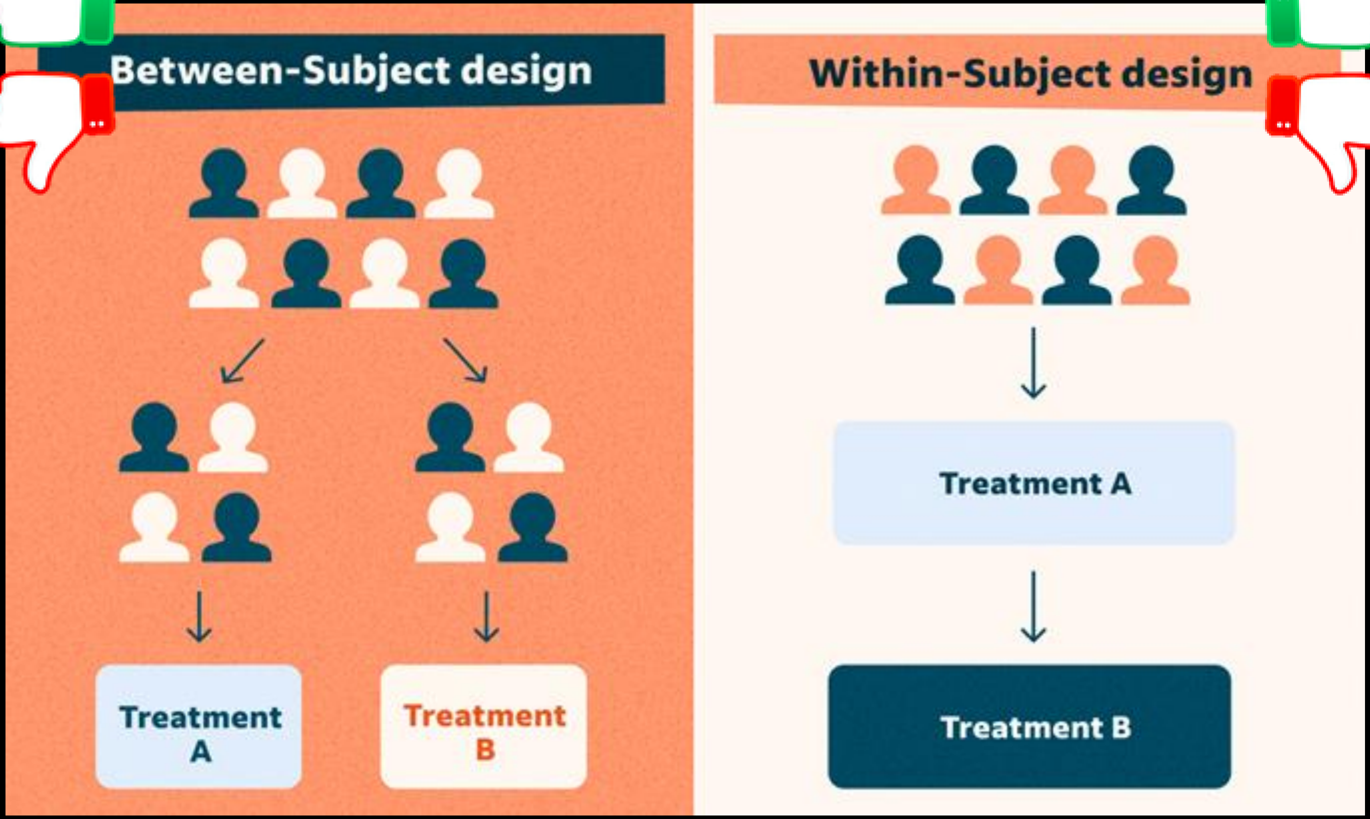
- High variability in means requires more participants



Reporting participants

- Sampling method
- Sample size
- Average age and gender split
- Recruitment process
- Participants’ background
- Other things if needed for specific study

Experimental design



independent-measures design

repeated-measures design

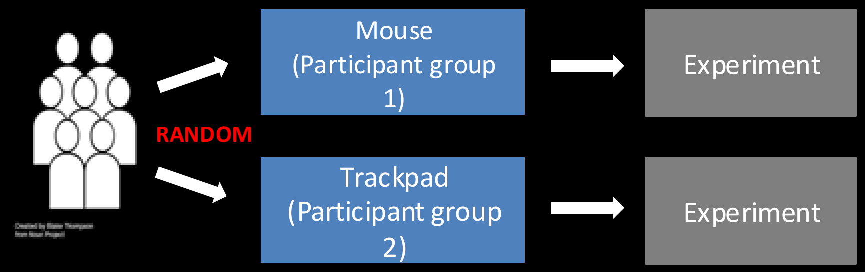
Between-Group Design

Advantages

- Simplicity
- Less chance of practice or fatigue effects
- Useful when it is impossible for an individual to participate in all conditions

Disadvantages

- Expense (time, effort, and number of participants)
- Individual differences (different responses to experimental manipulations)



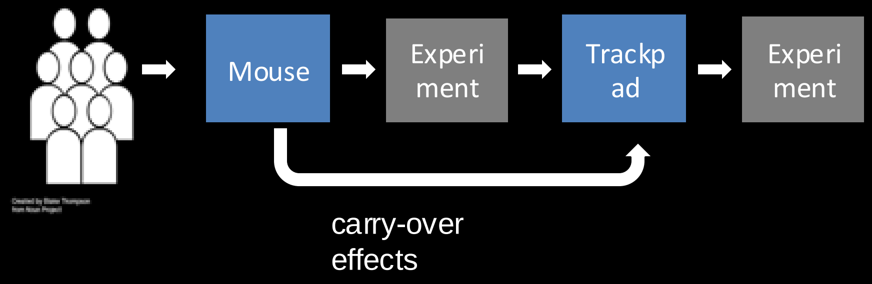
Within-Group Design

Advantages

- Economy
- Sensitiveness
- Cancelling out individual differences

Disadvantages

- Carry-over effects from previous conditions
- Conditions need to be reversible



Longitudinal studies

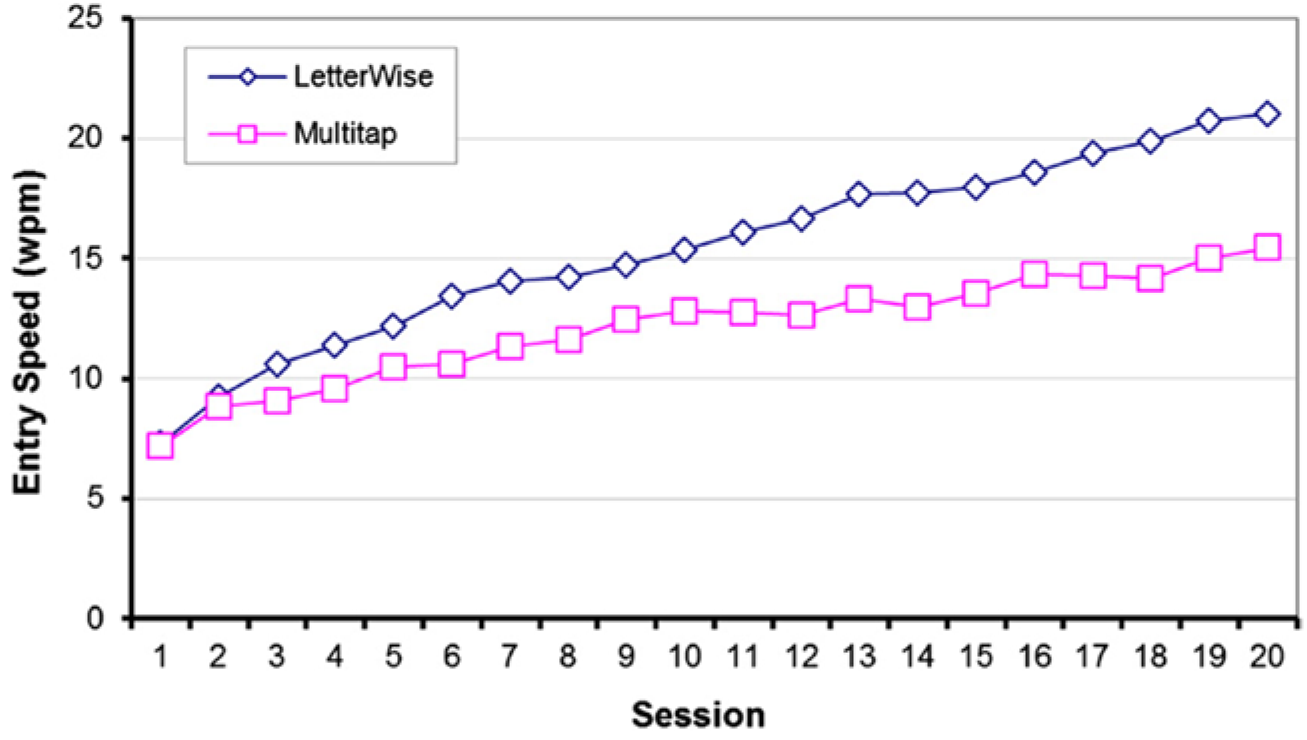
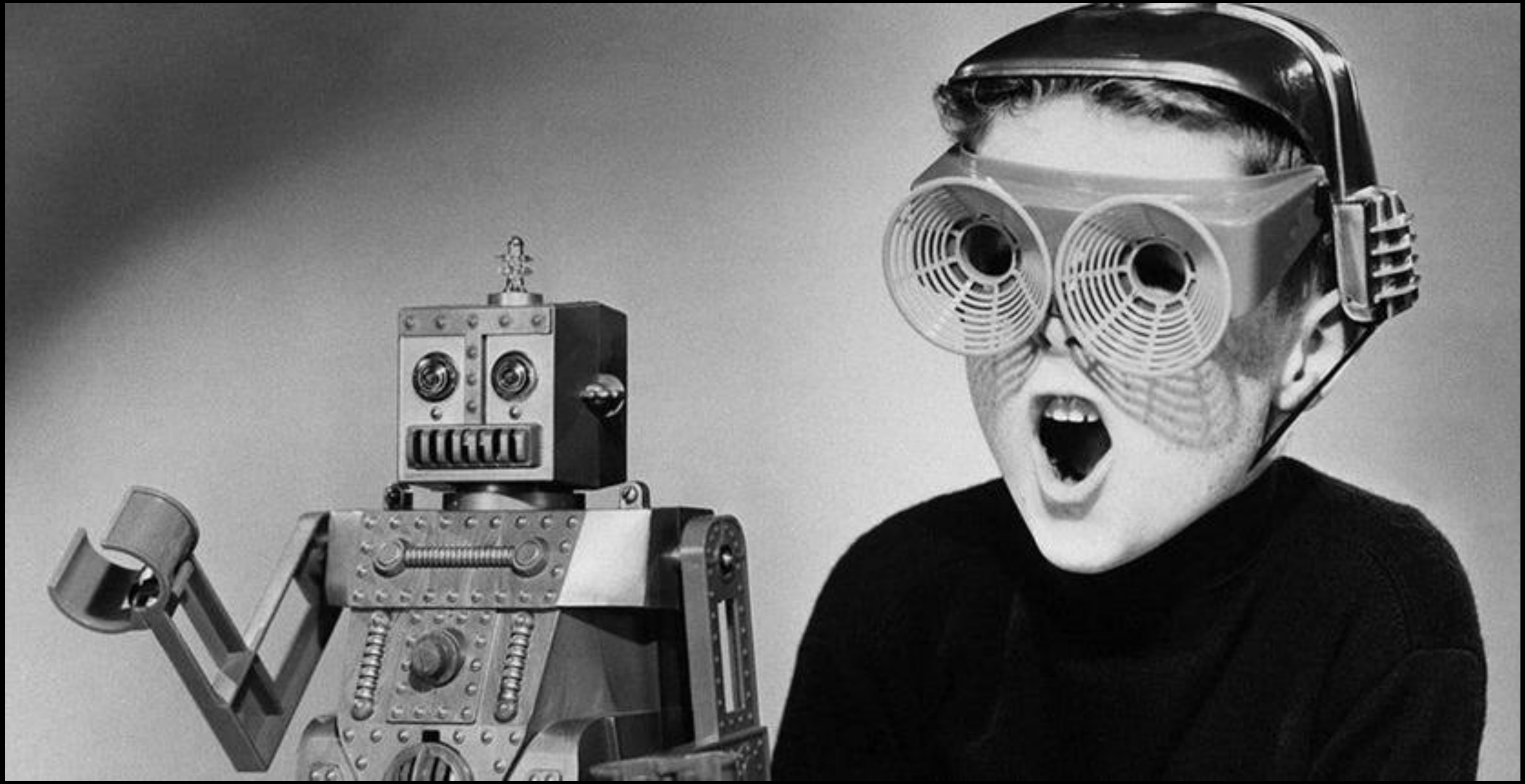


FIGURE 5.16

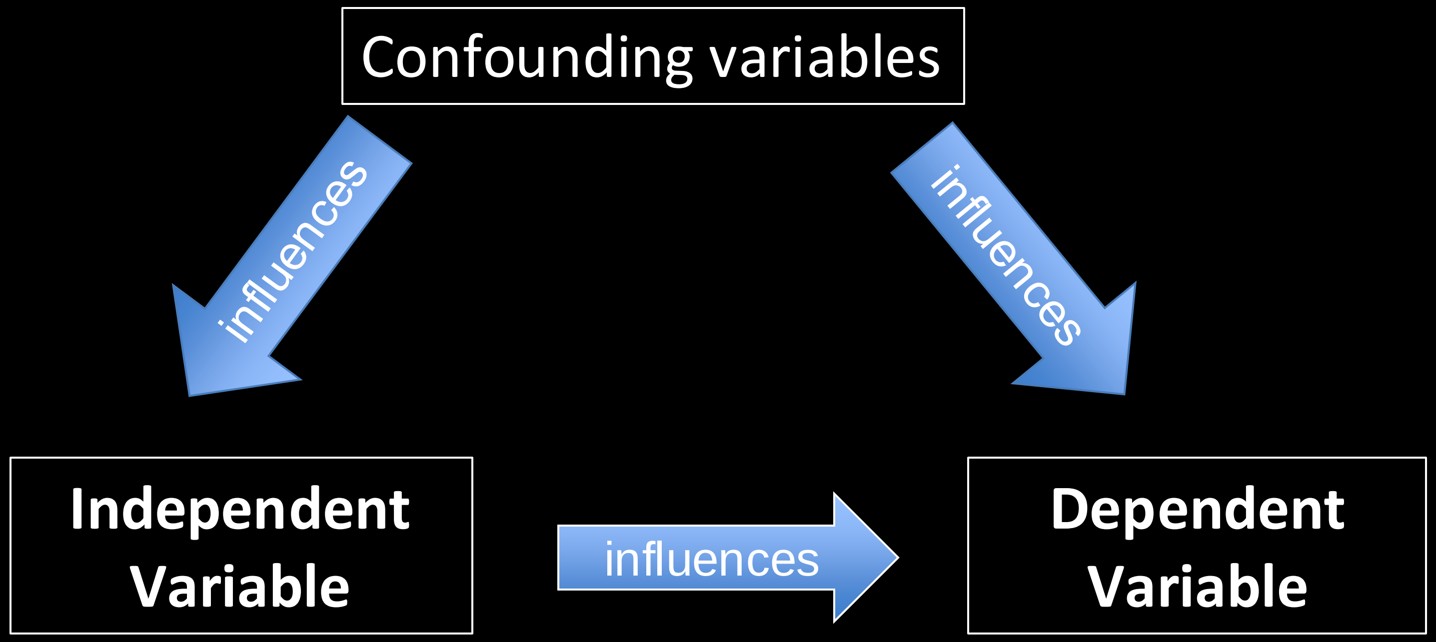
Example of a longitudinal study. Two text entry methods were tested and compared over 20 sessions of input. Each session involved about 30 minutes of text entry.

Novelty effect



Identify Confounding Variables

- Independent Variables
- Dependent Variables
- Uncontrolled variables (confounding variables)



Identify Confounding Variables

How to identify confound variables?

- Think about possibilities
  - What else could influence the results?
- Review the literature
  - What is already known?
- Experience
  - Talk to colleagues and supervisors

Control Confounding Variables

How to control confound variables?

- Randomize
- Record
  - Analyze it later as co-variable

If confound variables are not controlled, the recorded data can be unusable.

Randomization

How to manipulate a single aspect only?

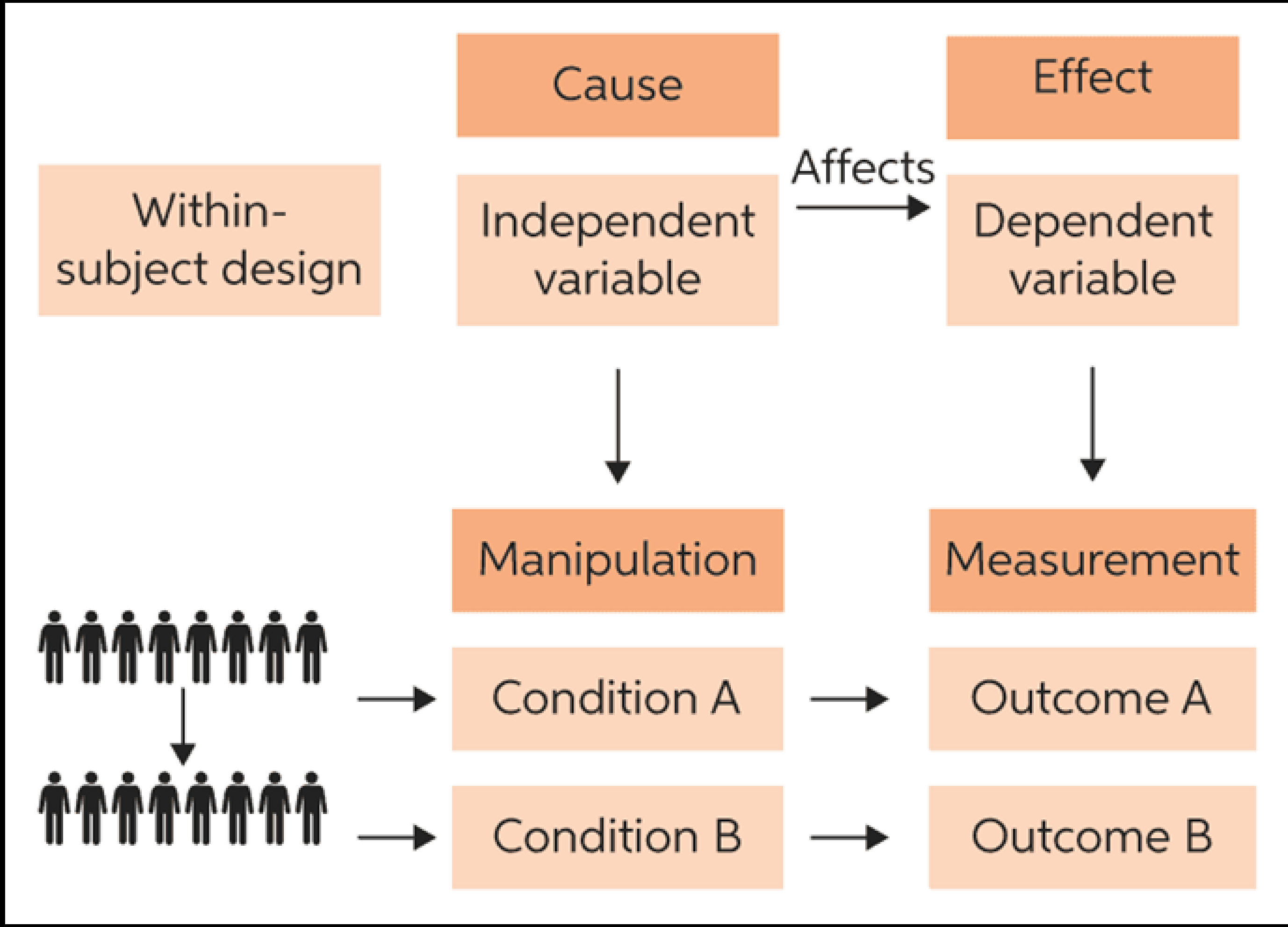
Approach: randomization

- Spread participants randomly into groups
  - Spreads intelligent, dumb, tired, excited, … people equally by chance
- Run conditions and tasks in random order
  - Avoid sequence effects, e.g. training or tiredness

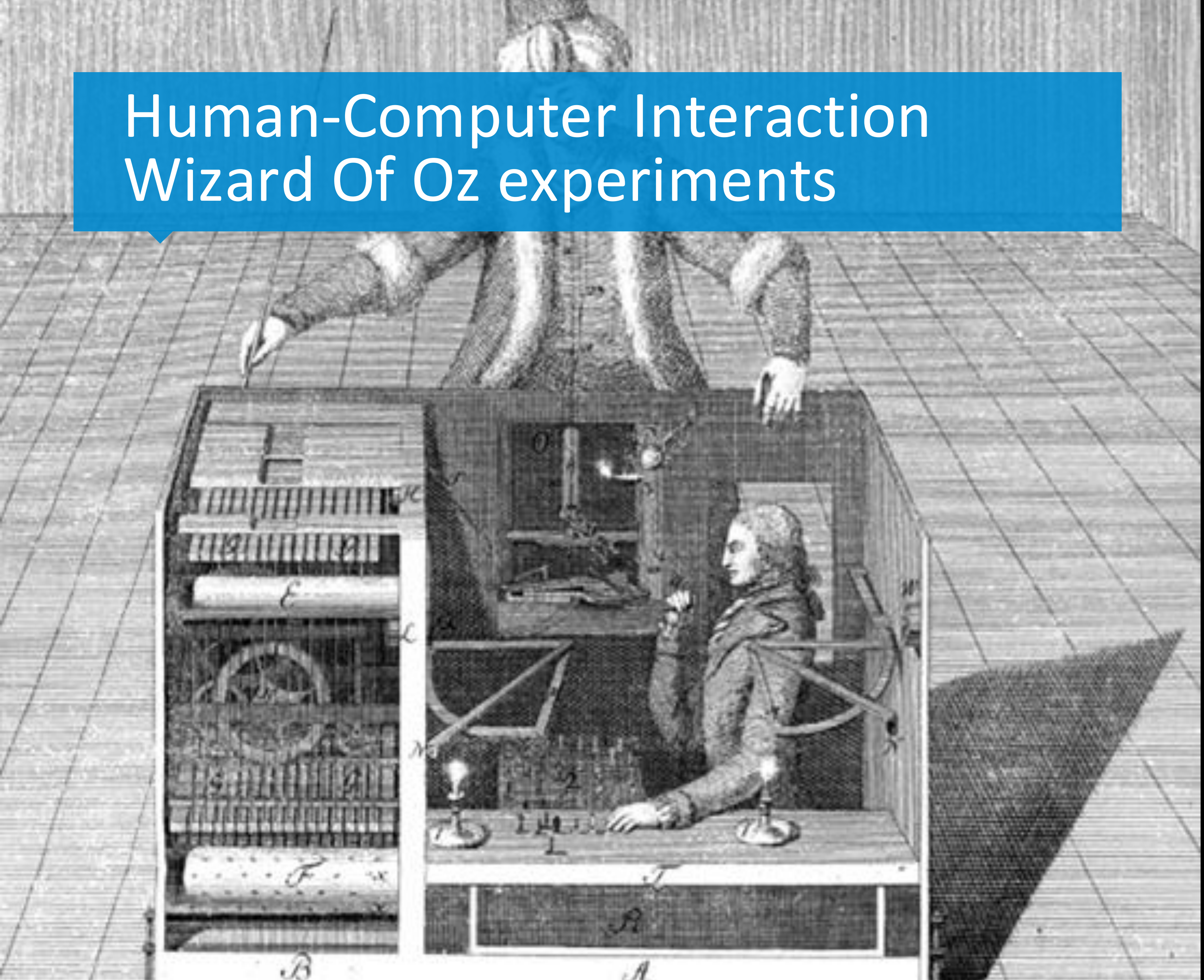
A	B	C
B	C	A
C	A	B

https://en.wikipedia.org/wiki/Latin_square

Randomization in within-subject design



Human-Computer Interaction
Wizard Of Oz experiments

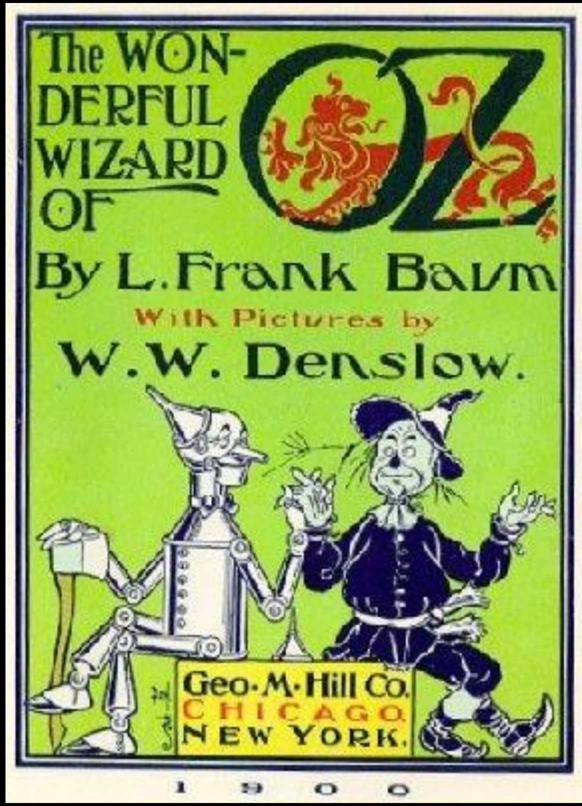


Development is hard – Testing is harder

- Often unclear if a system is worth the development effort
- Especially true for systems requiring novel hardware or algorithms
- Learning if the system is useful and the functions users want requires the system

Wizard of Oz - Definition

- "Wizard of Oz is a rapid-prototyping method for systems costly to build or requiring new technology. A human "Wizard" simulates the system's intelligence and interacts with the user through a real or mock computer interface."



Maulsby, D., Greenberg, S., & Mander, R. (1993). Prototyping an intelligent agent through Wizard of Oz.
In Proceedings of the INTERACT'93 and CHI'93 conference on Human factors in computing systems.

Wizard of Oz

- An invisible ‘wizard’ controlling parts of the functionality
- We only implement the easy parts but leave the hard part to the human operator
- Provides the user with the experience without extensive implementation effort for the prototype
- Typical areas
  - Speech recognition
  - Speech synthesis
  - Reasoning
  - Computer vision

Advantages and Disadvantages

Wizard of Oz

- Implemented only the “soft“ part
- Using human operators for the hard parts

Advantages

- Receive early feedback
- Compare with alternatives

Disadvantages

- Even implementing the soft parts requires effort
- Unclear if operator’s performance can ever be reached
- Performance depends on the operator

End of Presentation

Questions?